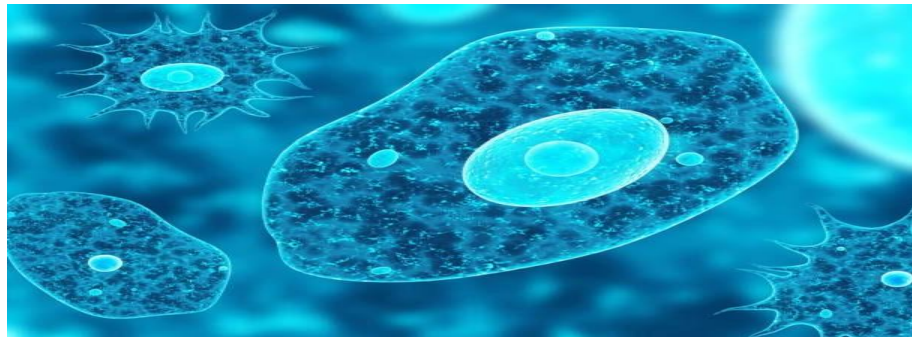




Cell Biology

CHROMOSOMAL DISORDERS



Lecture 9

Dr. Sally Atef

CLOs

- A3 Recognize information about genetic codes in transcription , translation and mutation
- C1 Interpret principles of cell structures, cell signaling, gene regulation, protein synthesis and mutation in treatment of disease.
- D1 Create proper scientific presentations using modern technology and media.

Lecture outlines

- Single gene disorder
 - 1. Autosomal Dominant
 - 2. Autosomal Recessive
 - 3. X-Linked Recessive
 - 4. X-Linked Dominant
- Chromosomal disorder

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graph TD; A[Gene and chromosomal disorders] --> B[Single Gene disorder]; A --> C[Chromosomal disorder]
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**Gene and
chromosomal
disorders**

**Single Gene
disorder**

**Chromosomal
disorder**

Single gene disorder

- also known as a **monogenic disorder**
- is a genetic condition caused by a **change or mutation in a single gene**.
- These mutations can be small, like a change in a single DNA base pair, or larger, like the deletion of an entire gene.

Single gene disorder

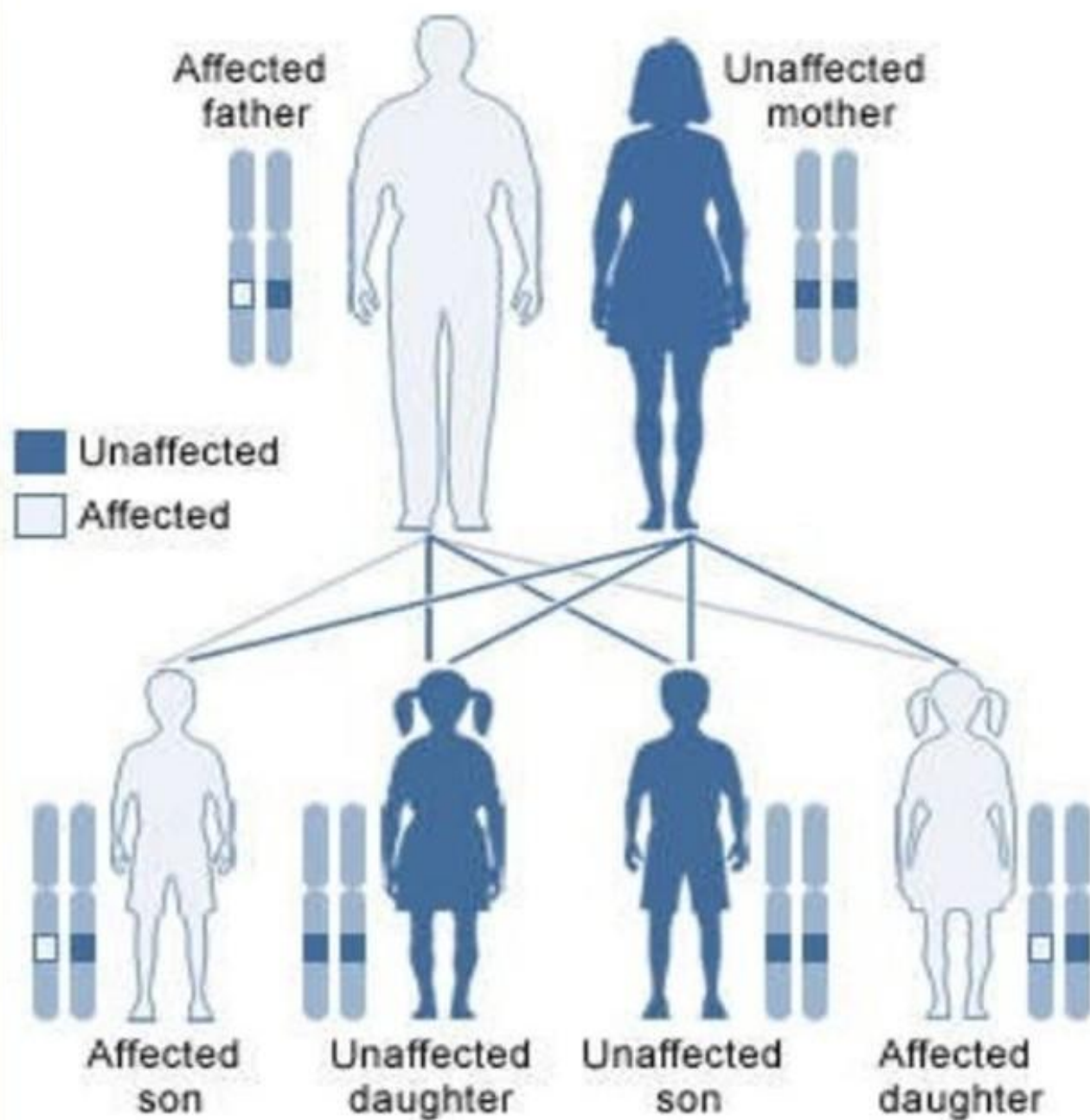
There are four main patterns:

- 1. Autosomal Dominant**
- 2. Autosomal Recessive**
- 3. X-Linked Recessive**
- 4. X-Linked Dominant**

1. Autosomal Dominant

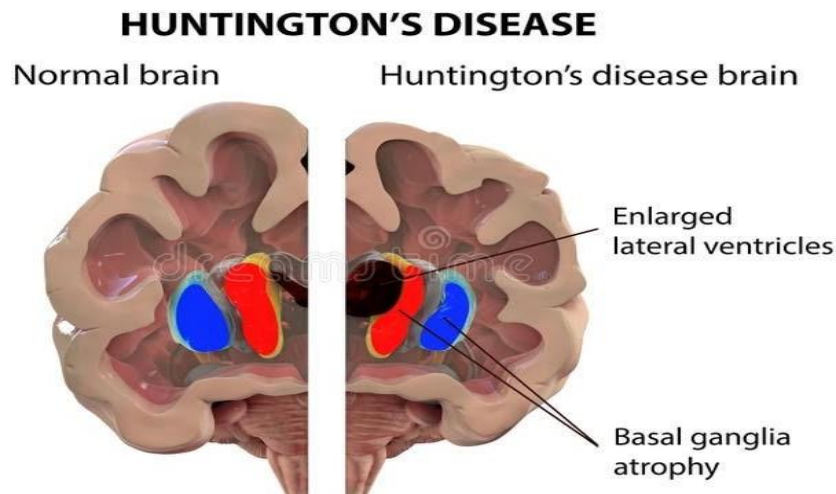
- **Gene Location:** An autosome (chromosomes 1-22).
- **Mutation Effect:** Only one copy of the mutated gene (from either parent) is needed to cause the disorder.
- **Family Pattern:**
 - An affected person typically has **one affected parent**.
 - The disorder often appears in **every generation**.
 - Each child of an affected parent has a **50% chance** of inheriting the mutated gene and the disorder.
- **Examples:**
 - **Huntington's Disease:** A neurodegenerative disorder.
 - **Marfan Syndrome:** A disorder affecting connective tissue.

Autosomal dominant



Huntington's Disease

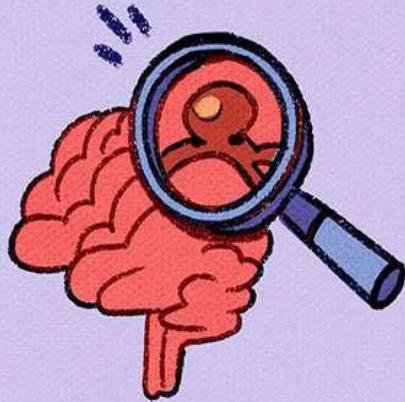
- is the degeneration of neurons in specific areas of the brain, particularly the **basal ganglia** and **cerebral cortex**, which are critical for movement control, thought, and emotion.
- A triad of **movement disorders** (especially chorea), **cognitive decline** (dementia), and **psychiatric/behavioral problems**.



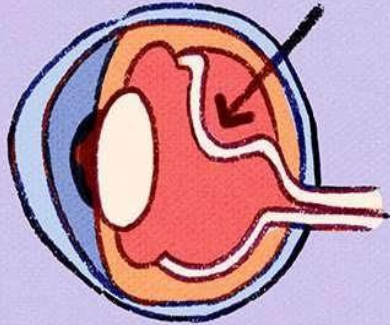
Marfan Syndrome

- a defect in the connective tissue makes it weaker and less elastic
- the syndrome can affect multiple organ systems, most notably the **skeleton, eyes, and heart and blood vessels**
- **Common symptoms are extremely long arms, fingers and legs, breastbone outward, long and crowded teeth, weak sight and flat feet**

Marfan Syndrome Symptoms



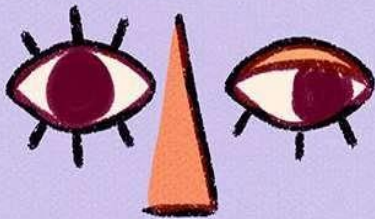
Aneurysm



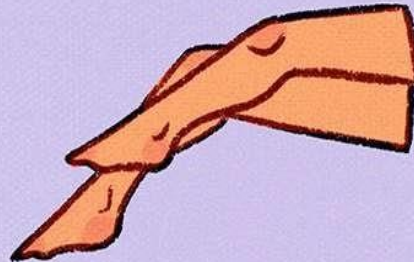
Retinal detachment



Joint pain



Lazy eye



Uneven leg length



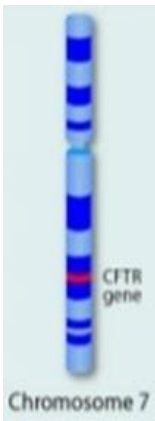
Chronic sinusitis

2. Autosomal Recessive

- **Gene Location:** An autosome.
- **Mutation Effect:** Two copies of the mutated gene (one from each parent) are needed to cause the disorder. A person with one copy is a **carrier**; they usually don't show symptoms but can pass the mutation to their children.
- **Family Pattern:**
 - **Parents** of an affected child are typically **carriers** and unaffected.
 - Each child of two carrier parents has a **25% chance** of being affected.
- **Examples:**
 - **Cystic Fibrosis:** A disorder affecting the lungs and digestive system.
 - **Sickle Cell Anemia:** A blood disorder affecting red blood cells.
 - **Galactosemia**

Cystic Fibrosis

- **autosomal recessive** genetic disorder that primarily affects the **lungs and digestive system**.
- It causes the production of abnormally thick and sticky mucus, which leads to severe **respiratory and gastrointestinal problems**.
- A mutation leads to a **defective CFTR protein**. Cystic Fibrosis Transmembrane Conductance Regulator (CFTR) protein is an essential ion channel on the surface of epithelial cells that moves chloride, bicarbonate, and water into and out of cells, maintaining thin, free-flowing mucus.



A Organs affected by cystic fibrosis

Sinuses:

sinusitis (infection)

Lungs: thick, sticky mucus buildup, bacterial infection, and widened airways

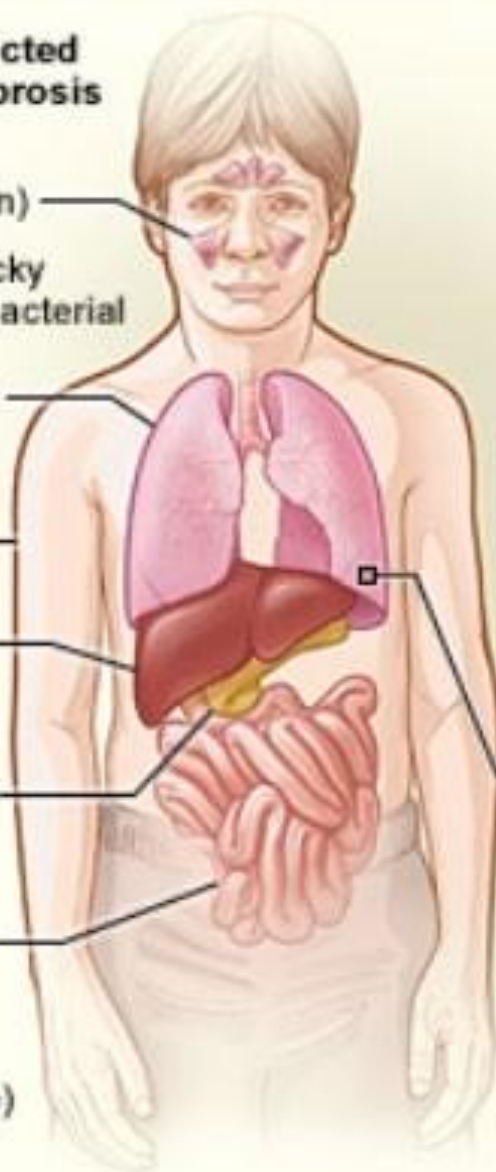
Skin: sweat glands produce salty sweat.

Liver: blocked biliary ducts

Pancreas: blocked pancreatic ducts

Intestines: cannot fully absorb nutrients

Reproductive organs: (male and female) complications



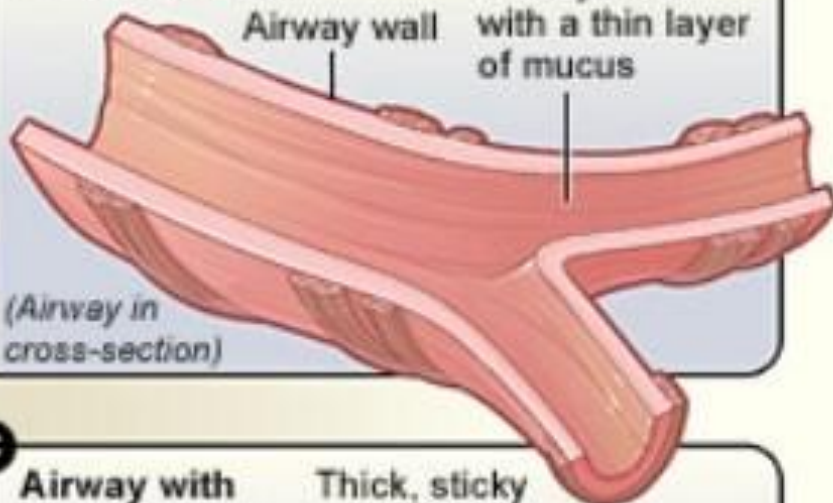
B

Normal airway

Airway wall

Airway lined with a thin layer of mucus

(Airway in cross-section)



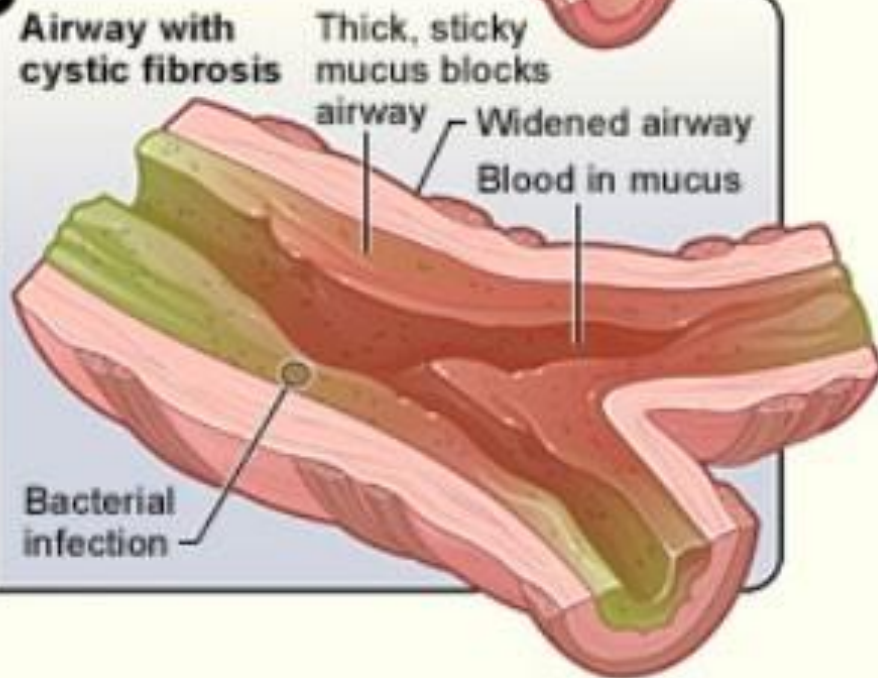
C

Airway with cystic fibrosis

Thick, sticky mucus blocks airway

Widened airway
Blood in mucus

Bacterial infection

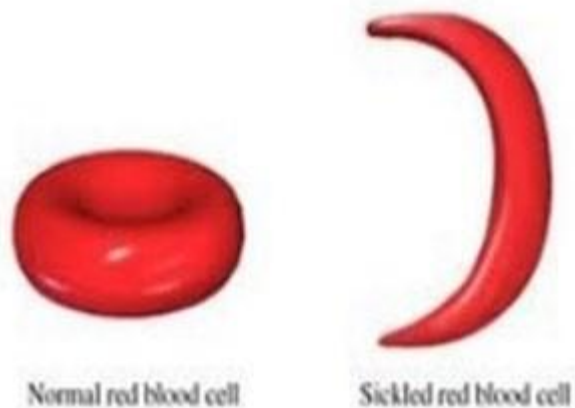
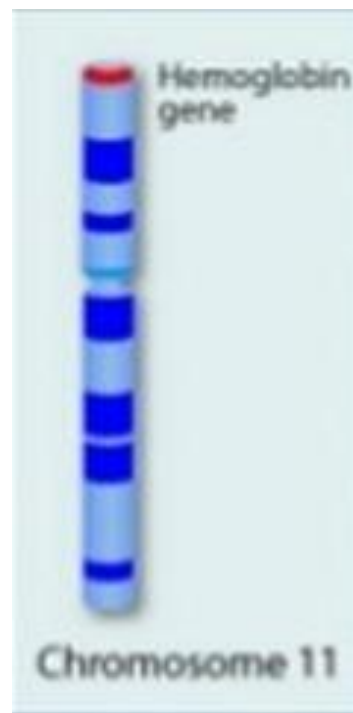
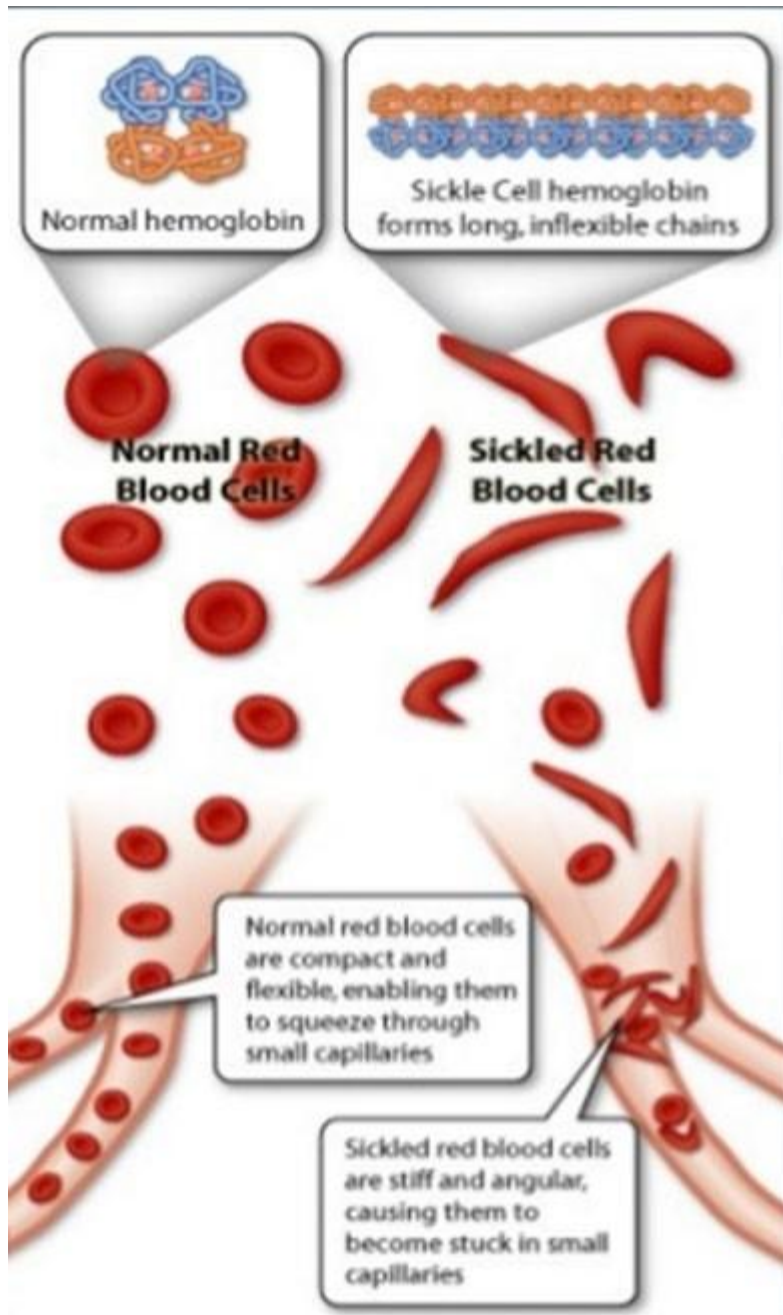


Galactosemia

- is a rare, inherited **metabolic disorder** that prevents the body from properly processing the sugar **galactose**. Galactose is a component of **lactose**, the sugar found in all mammalian milk
- Individuals with galactosemia lack a crucial enzyme needed to break down galactose, leading to a toxic buildup of galactose and its byproducts in the body, which causes severe damage.
- This buildup damages vital organs, particularly the **liver, kidneys, and brain**, and causes **cataracts** in the eyes.

Sickle cell disease

- a genetic mutation causes the body to produce abnormal hemoglobin, called **hemoglobin S (HbS)** causing **RBCs to become stiff, sticky, and shaped like crescent moons**
- **Hemolytic Anemia:** The sickle cells are fragile leading to a chronic shortage of red blood cells.
- **Vaso-occlusion:** The stiff, sticky sickle cells get stuck in small blood vessels, blocking blood flow. This prevents oxygen from reaching tissues, causing severe pain and organ damage.
- **Ischemia & Infarction:** The blocked blood flow (ischemia) can lead to tissue death



3. X-Linked Recessive

- **Gene Location:** The X chromosome.
- **Mutation Effect:** The mutation is on the X chromosome.

Males (XY) are more frequently and severely affected because they have only one X chromosome. **A single mutated copy will cause the disorder.**

Females (XX) have two X chromosomes and are usually carriers unless both copies are mutated.

- **Family Pattern:**
 - The disorder is typically passed from a **carrier mother** to her **son**.
 - Affected males do not pass the disorder to their sons (they pass their Y chromosome), but all their daughters will be carriers.
- **Examples:**
 - **Hemophilia A:** A disorder impairing blood clotting.
 - **Red-Green Color Blindness.**

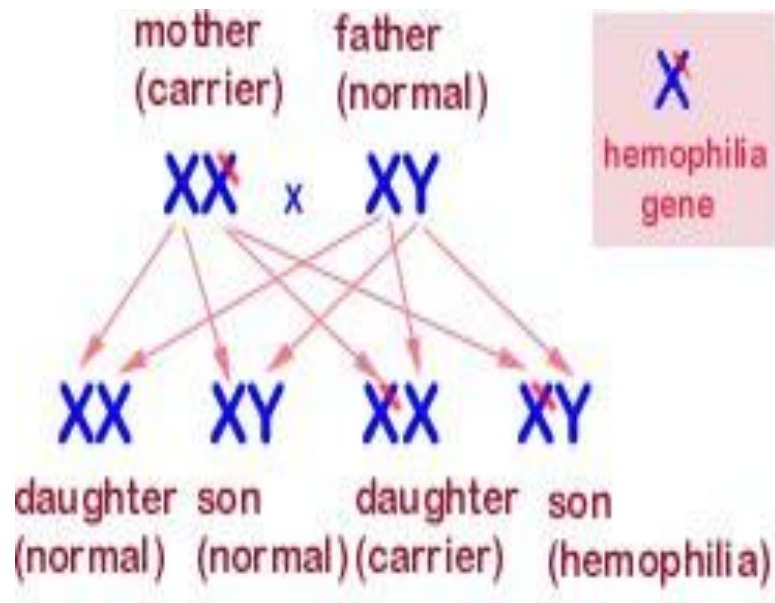
X-Linked Disorders

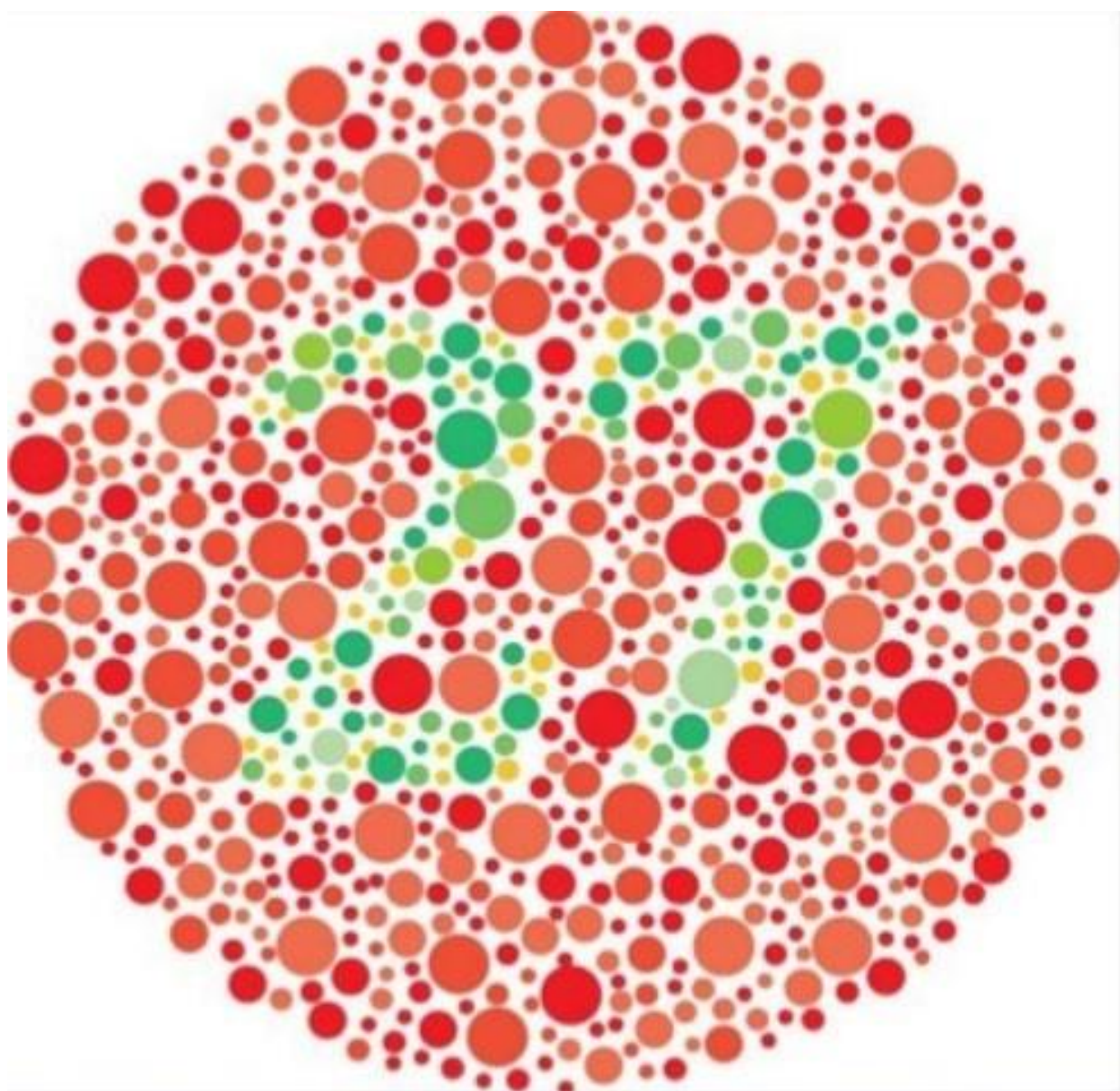
- X-Linked Recessive Inheritance:
- Characteristic Features:
 - Affected male $\rightarrow$ Sons unaffected, daughters are carriers
 $\text{xY} + \text{XX} \rightarrow \text{xX} \quad \text{xX} \quad \text{XY} \quad \text{XY}$
 - Heterozygous women $\rightarrow$ Sons 1 chance in 2 of disease
 $\text{XY} + \text{Xx} \rightarrow \text{XX} \quad \text{Xx} \quad \text{XY} \quad \text{xY}$

Hemophilia

A disorder in which a person's blood does not clot properly.

It is a recessive sex-linked, X-chromosome disorder.





Color blindness

- Individuals are unable to distinguish the shades of **red** and **green**
- The genes for the red and green cone pigments are located on the **X chromosome**.
 - **Males (XY)** have only one X chromosome. If that single X chromosome carries a defective colour vision gene, they will be colour blind.
 - **Females (XX)** have two X chromosomes. To be colour blind, both X chromosomes must carry the defective gene. If only one X chromosome has the defect, they are typically **carriers** with normal colour vision but can pass the trait to their children.
- This is why colour blindness is far more common in men.

4. X-Linked Dominant

- **Gene Location:** The X chromosome.
- **Mutation Effect:** A single mutated copy on the X chromosome will cause the disorder in both males and females. These disorders are much rarer.
- **Family Pattern:**
 - Affected males pass the disorder to all their daughters but none of their sons.
 - Affected females have a 50% chance of passing it to any child.
- **Example:**
 - **Fragile X Syndrome:** A leading cause of inherited intellectual disability.

Fragile X Syndrome

- inherited form of intellectual disability. It is caused by a mutation in a single gene on the X chromosome that disrupts its normal function, leading to a range of **developmental, cognitive, and behavioral challenges**.
- **Fragile X Mental Retardation Protein (FMRP)** is essential for normal brain development regulating the production of other proteins that are critical for learning, memory, and cognitive function.
- Without FMRP, this process is dysregulated, leading to the neurological and behavioral symptoms of FXS.

Fragile X syndrome

Symptoms can be found in the major categories below.

Some intelligence issues include:



Low intelligence quotient (IQ).



Delay of nonverbal communication.



Problems with math.

Some mental health problems include:



Anxiety.



Depression.



Obsessive compulsive behaviors.

Some physical features include:



A long, narrow face.



A large forehead.



Crossed eyes.

Some behavioral issues, like:



Social anxiety and shyness.



Attention-deficit/hyperactivity disorder (ADHD).

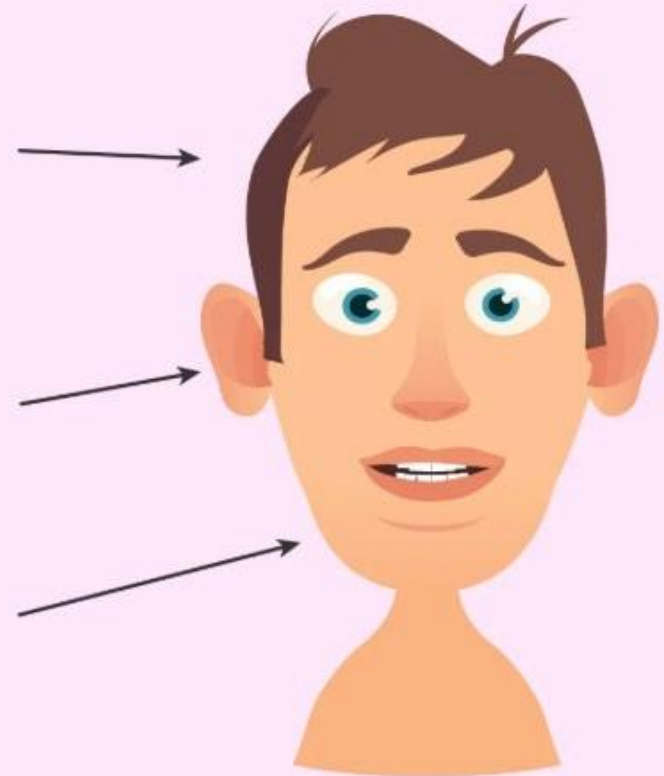


Poor eye contact.

Elongated face
and broad
forehead

Large,
protruding ears

Prominent
lower jaw



Chromosomal disorder

- A **chromosomal disorder** is a genetic condition caused by an abnormality in the **number or structure of chromosomes**.
- **Numerical abnormalities**
- **Structural abnormalities**

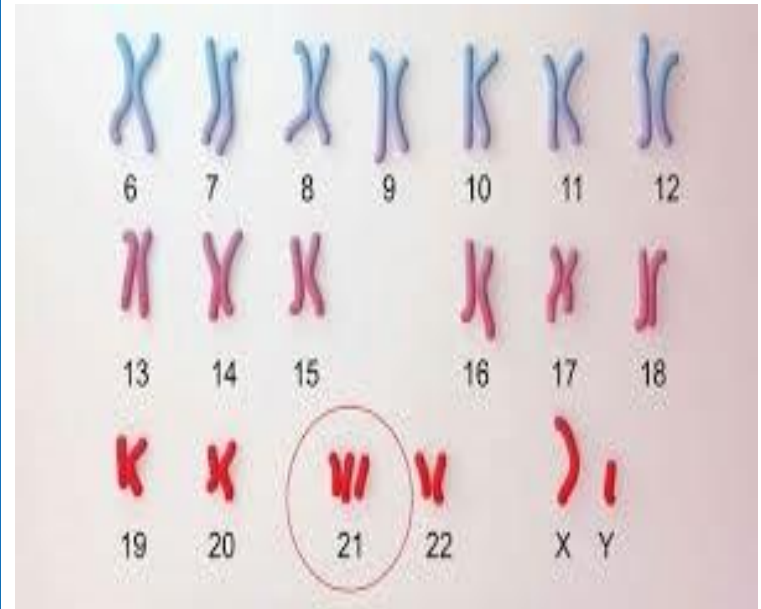
Main Types of Chromosomal Abnormalities

1. Numerical Abnormalities (Aneuploidy)

- This occurs when an individual has either an extra or a missing chromosome. This happens due to an error in cell division, where chromosomes fail to separate properly during the formation of sperm or egg cells.
- **Trisomy:** The presence of an **extra chromosome** (three copies instead of the usual two).
 - **Down Syndrome (Trisomy 21):** An extra copy of chromosome 21.
 - **Edwards Syndrome (Trisomy 18):** An extra copy of chromosome 18.
 - **Patau Syndrome (Trisomy 13):** An extra copy of chromosome 13.
 - **Klinefelter Syndrome (XXY):** An extra X chromosome in males.
 - **Triple X Syndrome (XXX):** An extra X chromosome in females.
- **Monosomy:** The **absence of one chromosome** (one copy instead of the usual two). Most autosomal monosomies are incompatible with life.
 - **Turner Syndrome (X):** Females with only one X chromosome (X instead of XX).

Down syndrome

- This syndrome is a type of trisomy as there is an extra copy of **chromosome 21**.



Edward syndrome

caused by the presence of an **extra copy of chromosome 18**. It is characterized by severe intellectual disability and multiple life-threatening medical complications.

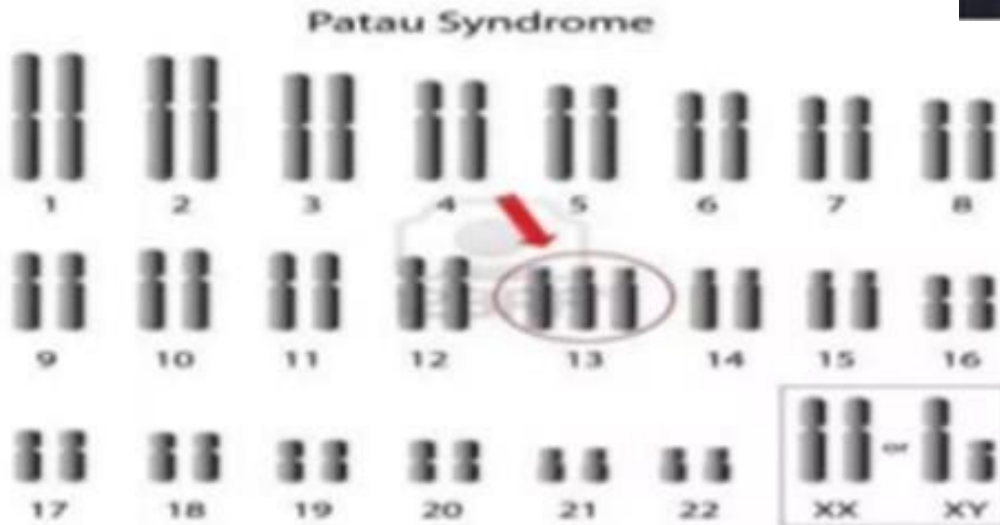
5-10% of infants live past their first year. Those who survive are almost always female and have profound intellectual and physical disabilities.



Patau Syndrome

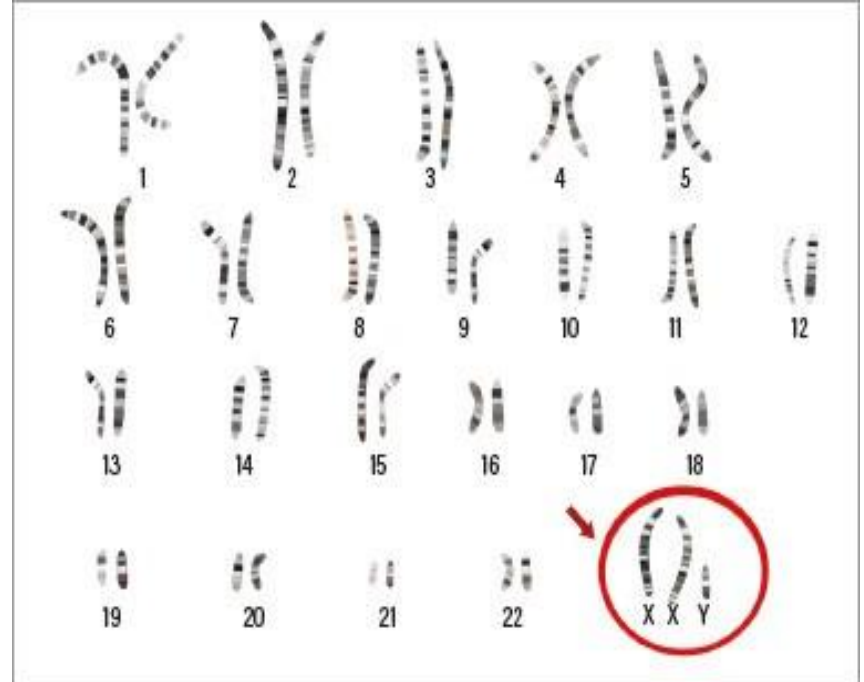
An extremely severe condition with profound **brain, facial, and organ malformations**.

the vast majority of infants die in the first few weeks or months of life.



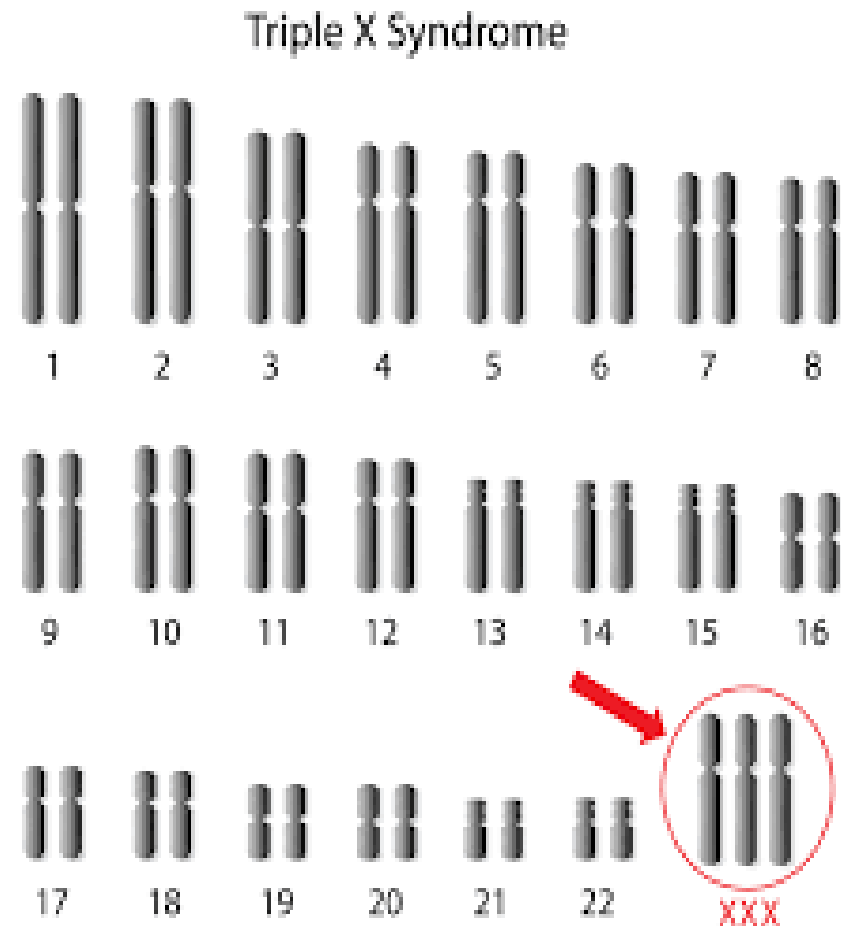
Klinefelter syndrome

- **Tall Stature:** Longer legs and arms, with a shorter torso.
- **Hypogonadism:**
 - Delayed or incomplete puberty.
 - Reduced facial, body, and pubic hair.
 - **Gynecomastia:** Breast tissue development
 - **Weakened Bones (Osteoporosis)**
- **Reduced Muscle Mass and Strength**
- **Infertility and Low Testosterone**



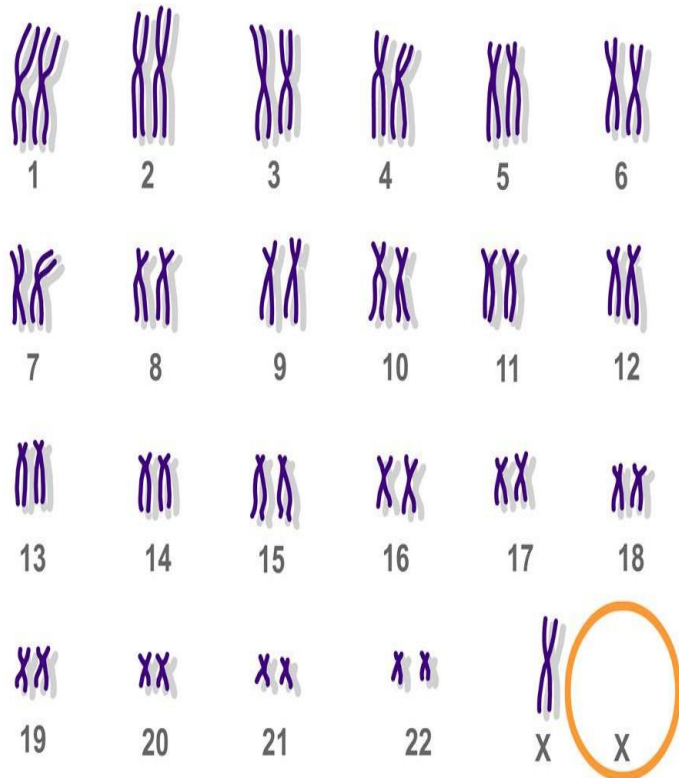
Trisomy X syndrome

- Symptoms include tall stature; speech delays, menstrual abnormalities, and weak muscle tone although many girls and women exhibit no symptoms.

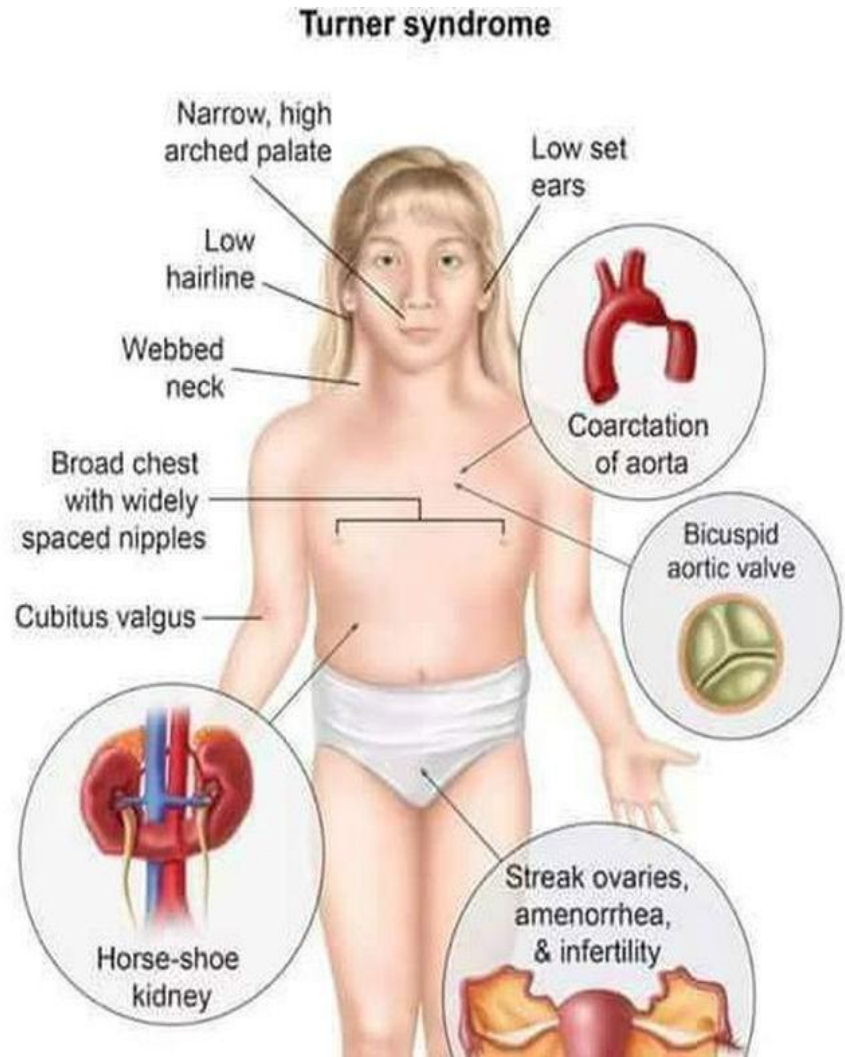


Monosomy

Turner Syndrome



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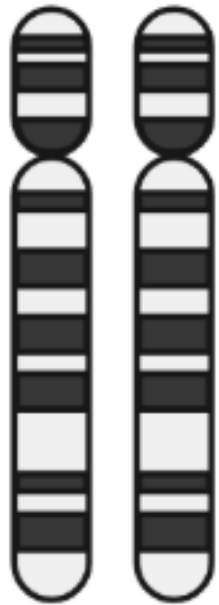
Main Types of Chromosomal Abnormalities

2. Structural Abnormalities

- This occurs when the structure of a chromosome is altered. Pieces of a chromosome can be broken and rearranged in various ways. These can be inherited from a parent or occur *de novo* (for the first time).
- **Deletions:** A piece of a chromosome is missing.
- **Duplications:** A piece of a chromosome is duplicated, resulting in extra genetic material.
- **Translocations:** A piece of one chromosome breaks off and attaches to another chromosome.
- **Inversions:** A segment of a chromosome is reversed end-to-end.

Cri du Chat Syndrome

- rare genetic disorder caused by a missing piece of chromosome 5. It is also known as **5p- syndrome** (5p minus syndrome).
- The symptoms and their severity vary depending on the size and location of the deletion.
- **The Cat-like Cry:** The most distinctive feature is high-pitched, weak, and monotone voice due to laryngeal hypoplasia



Normal
chromosome 5



Deletion on
chromosome 5

